Incident Report: Authentication Log Investigation

Cyber Security Internship, Task 04 | Analyst: [Your Name] | Log date: 2026-08-01 (UTC) | Data: synthetic

1. Executive summary

The log contains two attack patterns and one event that needs verification. **stu-117** was the target of a brute-force attempt and was locked out. Four accounts (**stu-209, stu-311, stu-415, stu-502**) were hit by a password-spraying run from **203.0.113.42**; every spray attempt failed. About 2 h 26 min later, **stu-311** logged in successfully from a new IP and device. This is **not proven** to be malicious, but because the account had just been sprayed it should be verified with the user. No confirmed account compromise is shown by the log.

2. Scope and method

Data: authentication-log-sample.csv, 12 events from 09:00:12 to 11:47:09 UTC, 6 accounts, 4 source IPs. **Normalization:** timestamps parsed to UTC date-times, text fields trimmed and lower-cased, events sorted by time and given IDs E01-E12 so each finding can cite its evidence. **Detection:** three rules, applied in the workbook (formulas) and in detect.py (code), which give identical results.

Rule	Logic	Result
1 Brute force	3 or more failed logins for one account from one IP within 5 min	1 match: stu-117 from 198.51.100.27
2 Password spraying	Failures from one IP against 3 or more accounts within 2 min	1 match: 203.0.113.42, 4 accounts in 12 s
3 Unusual access	Successful login from an IP not previously seen for the account, after earlier failures on it	1 review: E12, stu-311 from 192.0.2.99

3. Findings

#	Finding	Evidence	Severity / confidence
1	Brute-force attempt on stu-117. The account lock stopped it.	E02-E04: 3 failures from 198.51.100.27 (windows) in 34 s. E05: account_lock 45 s later. No later success.	Medium / Medium
2	Password spraying against 4 accounts. All attempts failed.	E06-E09: one failure each for stu-209, 311, 415, 502 from 203.0.113.42, device 'unknown', 12 s total. E10: rate_limit recorded for stu-209 only.	High / High
3	Successful login on a previously sprayed account from a new IP and device.	E07: stu-311 failure from 203.0.113.42 at 09:21:13. E12: stu-311 success from 192.0.2.99 (ios) at 11:47:09.	Medium / Low (needs verification)

4. Timeline (UTC, 2026-08-01)

Time	Events	What happened
09:00:12	E01	stu-104 logs in from 192.0.2.18 (android). Baseline, normal.
09:03:44 - 09:04:18	E02-E04	3 failed logins for stu-117 from 198.51.100.27.
09:05:03	E05	stu-117 account locked.
09:21:10 - 09:21:22	E06-E09	Spray: 4 accounts fail from 203.0.113.42 in 12 s.
09:22:01	E10	rate_limit block recorded for stu-209.
10:15:32	E11	stu-104 refresh token from the same IP and device as E01. Normal.
11:47:09	E12	stu-311 logs in successfully from 192.0.2.99 (ios). First time this IP and device appear for the account.

5. Evidence versus assumptions

Established by the log (evidence)	Inferred or unverified (assumptions)
stu-117: 3 failures from one IP, then locked, no later success. 203.0.113.42 failed against 4 accounts in 12 s; no spray attempt succeeded. stu-311 had a failure at 09:21:13 and a success from a different IP at 11:47:09.	The stu-117 failures were an attack and not a mistyped password. The spray is scripted (inferred from 3-5 s spacing and unknown device). The E12 login is a compromise: **not established**. It may be the real user on a new network. The two attacker IPs are related: no evidence.

Observation: 192.0.2.99 is in the same /24 range as stu-104's IP (192.0.2.18). This weakly suggests a shared network, but it is not proof the logins are related.

6. Recommended response

When	Action
Immediately	Contact the owner of stu-311 and confirm the 11:47 login. If not recognised: revoke sessions and refresh tokens, force a password reset, and review activity after 11:47. Block or closely monitor 203.0.113.42; consider 198.51.100.27 as well. Notify owners of stu-209, 311, 415, 502 and stu-117, and check them for unexpected activity.
Short term	Add IP-level rate limiting and a cross-account spraying alert (many accounts, one IP). Per-account lockouts alone did not stop the spray. Require MFA, and block passwords found in known breaches.
Longer term	Alert on a first-seen IP or device soon after an account was targeted. Log more fields (user agent, geolocation, reason for block) and keep longer history so each account has a baseline. Feed these rules into a central monitoring tool.

7. Limitations

Only 12 synthetic events from one day are available, with no geolocation, user-agent or history for most accounts. The log shows a rate_limit block for stu-209 but not whether the source IP itself was blocked. Findings are limited to what these events can support. Supporting files: investigation_workbook.xlsx (cleaned log, formulas, rules, timeline), detect.py and detect_output.txt (repeatable detection).